

# Mini Project Handout - Updated Group Allocation

## Practical 10: End-to-End DevOps Pipeline

Agile Software Development & DevOps Lab (20151119) | TE AI&DS | Duration: 2 Hours

### 1. Project Overview (Common for All Groups)

Every group builds a very small REST application for its own topic and takes it through a complete DevOps pipeline. The application is intentionally tiny (3 endpoints, data kept in memory, no real database) so that the full time goes into the pipeline, not into coding.

**Pipeline flow:** *Jira (Plan) -> Git/GitHub (Code) -> GitHub Actions (Build + Test) -> Docker (Package) -> Docker Hub (Release) -> Docker Compose (Deploy) -> Prometheus + Grafana (Monitor)*

**Application requirement (same for every topic):**

- GET /items : returns the list of items for the topic (e.g., tasks, books, students).
- POST /items : adds a new item to the list.
- GET /health : returns a simple 'OK' message (used for monitoring checks).

### 2. Time Plan for the 2-Hour Slot

| Time         | Activity                                                                                  |
|--------------|-------------------------------------------------------------------------------------------|
| 0 - 10 min   | Login to accounts, clone/open starter code, roles confirmed within the group.             |
| 10 - 25 min  | R1 sets up Jira (Epic, Stories, Sprint). R2 adapts the app to the topic in parallel.      |
| 25 - 45 min  | R2 pushes code through branch and Pull Request. R3 writes the CI workflow and Dockerfile. |
| 45 - 65 min  | CI run turns green. R3 builds image and pushes to Docker Hub.                             |
| 65 - 90 min  | Deployment and monitoring: docker-compose, Prometheus target UP, Grafana panel.           |
| 90 - 120 min | Demonstration and viva: about 5 minutes per group (max 6 groups per batch).               |

### 3. Individual Tasks by Role

Roles are assigned in the order the roll numbers appear in each group's list below (first listed roll = R1, second = R2, and so on). The number of roles depends on the group size.

#### Groups of 4 Members

| Role | Title                              | Individual Task                                                                                                                                                                                                                                   |
|------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1   | Agile Planner & Documentation Lead | Jira: create project, 1 Epic, 3 User Stories (view list, add item, health check), start a sprint and move cards to Done. Link the Jira issue key to a GitHub commit. Collect screenshots from all members and compile the group report.           |
| R2   | Developer & Version Control        | Take the starter Flask app and adapt it to the group's topic (3 endpoints, in-memory data) + 1 simple test. Create GitHub repo, work on a feature branch, raise a Pull Request and merge to main. Use the Jira issue key in every commit message. |
| R3   | CI/CD & Containerization Engineer  | Write the GitHub Actions workflow (checkout, install, run test) and show a green run. Write the Dockerfile, build the image, and push it to Docker Hub using GitHub Secrets.                                                                      |

|    |                                  |                                                                                                                                      |
|----|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| R4 | Deployment & Monitoring Engineer | Write docker-compose.yml to run app + Prometheus + Grafana together. Verify the Prometheus target is UP and build one Grafana panel. |
|----|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|

## Groups of 5 Members

| Role | Title                              | Individual Task                                                                                                                                                                                                                                   |
|------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1   | Agile Planner & Documentation Lead | Jira: create project, 1 Epic, 3 User Stories (view list, add item, health check), start a sprint and move cards to Done. Link the Jira issue key to a GitHub commit. Collect screenshots from all members and compile the group report.           |
| R2   | Developer & Version Control        | Take the starter Flask app and adapt it to the group's topic (3 endpoints, in-memory data) + 1 simple test. Create GitHub repo, work on a feature branch, raise a Pull Request and merge to main. Use the Jira issue key in every commit message. |
| R3   | CI/CD & Containerization Engineer  | Write the GitHub Actions workflow (checkout, install, run test) and show a green run. Write the Dockerfile, build the image, and push it to Docker Hub using GitHub Secrets.                                                                      |
| R4   | Deployment Engineer                | Pull the pushed image from Docker Hub and run the app using docker-compose.yml. Verify all endpoints in the browser. Bonus: deploy the same image on Minikube.                                                                                    |
| R5   | Monitoring & Security Engineer     | Configure Prometheus (prometheus.yml) and Grafana dashboard with one panel. DevSecOps check: confirm no passwords/keys are in code (only GitHub Secrets used).                                                                                    |

## 4. Batch-wise Group Allocation (Updated, Random Shuffle)

### Batch B3 (Roll No. 51 - 67)

| Group | Project Topic      | Members and Roles                                                                                                                                                                                  |
|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B3-G3 | Employee Directory | Roll 67: R1 - Agile Planner & Documentation Lead<br>Roll 56: R2 - Developer & Version Control<br>Roll 60: R3 - CI/CD & Containerization Engineer<br>Roll 61: R4 - Deployment & Monitoring Engineer |

## 5. Deliverables

### Group submission (one per group):

- GitHub repository link (with commit history, branch and merged Pull Request).
- Screenshots: Jira board/sprint, green GitHub Actions run, Docker Hub image, running app in browser, Prometheus target UP, Grafana dashboard.
- Short group report (5 to 8 pages): topic, architecture/pipeline diagram, tools used, steps, challenges, conclusion.

### Individual submission (in each student's lab journal):

- Practical 10 entry in the usual format: Aim, brief theory, project title, own role and task performed, key screenshots of own work, conclusion.
- Attach a copy of the group report.

## 6. Suggested Evaluation (10 Marks)

| Criteria                                                     | Marks | How Assessed                |
|--------------------------------------------------------------|-------|-----------------------------|
| Working end-to-end pipeline (commit to monitored deployment) | 4     | Live group demo             |
| Individual role performance and understanding                | 3     | Viva question on own role   |
| Documentation (group report + journal entry)                 | 2     | Submitted files             |
| Teamwork and time management                                 | 1     | Observation during the slot |

***Suggested viva questions by role:*** R1 - difference between Epic and Story; R2 - what is a Pull Request and why use branches; R3 - what does the workflow file do and why use Secrets; R4 - purpose of docker-compose; R5 - what is a Prometheus exporter/scrape target.